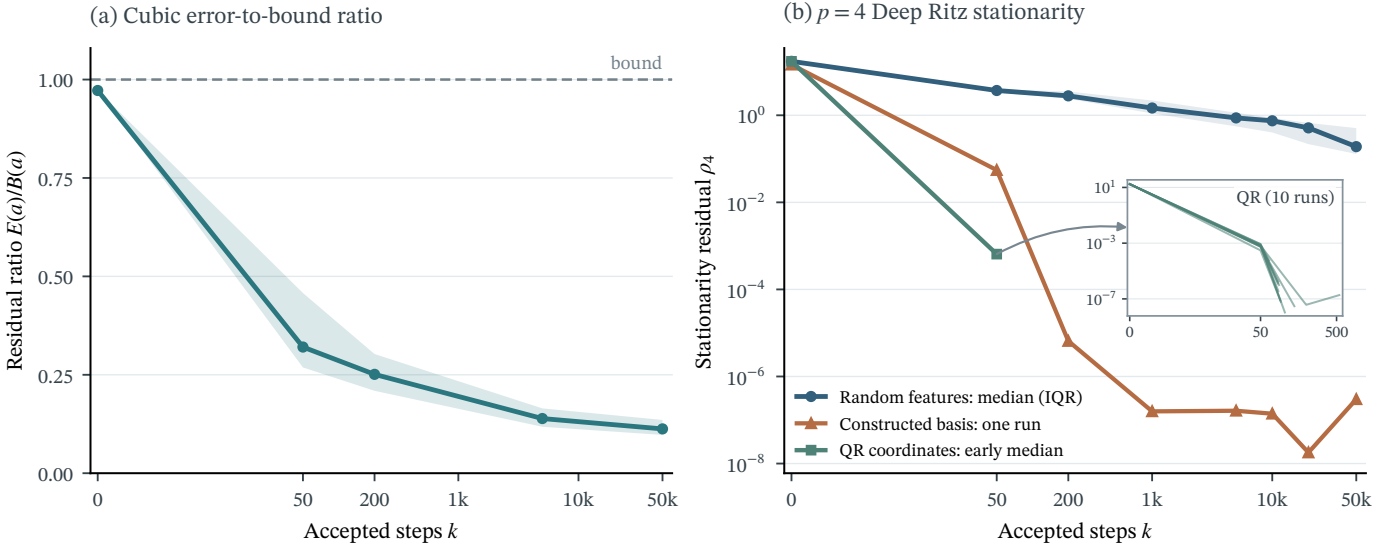
(a) Cubic error-to-bound ratio
bound
1.00
0.75
0.50
0.25
0.00
Residual ratio $E(a)/B(a)$
0
50
200
1k
10k
50k
Accepted steps k
(b) $p = 4$ Deep Ritz stationarity
Stationarity residual ρ_4
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
QR (10 runs)
10^{1}
10^{-3}
10^{-7}
0
50
500
Random features: median (IQR)
Constructed basis: one run
QR coordinates: early median
0
50
200
1k
10k
50k
Accepted steps k